Guth and Tye were interested in this possibility because their calculations suggested it might be relevant to a problem (the *magnetic monopole* problem[8]) researchers had encountered with various attempts at grand unification. But Guth and Tye realized that there might be another implication and, in retrospect, that's why their work proved pivotal. They suspected that the energy associated with a supercooled Higgs field— remember, the height of the field represents its energy, so the field has zero energy only if its value lies in the bowl's valley— might have an effect on the expansion of the universe. In early December 1979, Guth followed up on this hunch, and here's what he found.

A Higgs field that has gotten caught on a plateau not only suffuses space with energy, but, of crucial importance, Guth realized that it also contributes a uniform *negative pressure.* In fact, he found that as far as energy and pressure are concerned, a Higgs field that's caught on a plateau has the same properties as a cosmological constant: it suffuses space with energy and negative pressure, and in exactly the same proportions as a cosmological constant. So Guth discovered that a supercooled Higgs field does have an important effect on the expansion of space: like a cosmological constant, it exerts a repulsive gravitational force that drives space to expand.[9]

At this point, since you are already familiar with negative pressure and repulsive gravity, you may be thinking, All right, it's nice that Guth found a specific physical mechanism for realizing Einstein's idea of a cosmological constant, but so what? What's the big deal? The concept of a cosmological constant had long been abandoned. Its introduction into physics was nothing but an embarrassment for Einstein. Why get excited over rediscovering something that had been discredited more than six decades earlier?

Inflation

Well, here's why. Although a supercooled Higgs field shares certain features with a cosmological constant, Guth realized that they are not completely identical. Instead, there are two key differences—differences that make all the difference.